

Project repository.

1 Problem statement

The goal of this work is to forecast the values of a time series using the econometric models covered in the course.

The ARIMA(p, d, q) model extends ARMA-type models to non-stationary time series that can be made stationary after differencing. The ARIMA(p, d, q) model for a series y_t is defined as an ARMA(p, q) model applied to the d -th order differences of y_t .

The **ARIMA(p, d, q)** model:

$$\Delta^d y_t = \alpha_1 \Delta^d y_{t-1} + \dots + \alpha_p \Delta^d y_{t-p} + \varepsilon_t + \beta_1 \varepsilon_{t-1} + \dots + \beta_q \varepsilon_{t-q},$$

where y_t is the value of the time series at time t , $\Delta^d = (1 - L)^d$, ε_t is white noise, and L is the lag operator.

The generalization of the ARIMA model to series with a seasonal component is called SARIMA. Let s be the known seasonality of the series. We add to the ARIMA(p, d, q) model the components responsible for values in previous seasons. The SARIMA model can then be written as follows:

The **SARIMA(p, d, q)(P, D, Q)_s** model:

$$\begin{aligned} \Delta_s^D \Delta^d y_t &= \alpha_1 \Delta_s^D \Delta^d y_{t-1} + \dots + \alpha_p \Delta_s^D \Delta^d y_{t-p} \\ &+ \varepsilon_t + \beta_1 \varepsilon_{t-1} + \dots + \beta_q \varepsilon_{t-q} \\ &+ \alpha_1^s \Delta_s^D \Delta^d y_{t-s} + \dots + \alpha_p^s \Delta_s^D \Delta^d y_{t-ps} \\ &+ \beta_1^s \varepsilon_{t-s} + \beta_Q^s \varepsilon_{t-Qs}. \end{aligned}$$

where $\Delta_s^D = (1 - L^s)^D$.

A further generalization of SARIMA is the SARIMAX model. SARIMAX (Seasonal AutoRegressive Integrated Moving Average with eXogenous inputs) is an econometric model for time-series forecasting that accounts for seasonality and external factors.

2 Methodology

2.1 Data

This work addresses forecasting the number of sales for a retail chain in Ecuador for 2016. The data were taken from Kaggle. Three files were used (`train.csv`, `stores.csv`, `oil.csv`), from which the final dataset was assembled.

Description of the variables in `train.csv`:

- *store_nbr* identifies the store where the goods are sold.
- *family* identifies the type of product sold.
- *sales* is the total sales for a given product group in a particular store on a given date. Fractional values are possible, since goods can be sold in fractional units (e.g., 1.5 kg of cheese rather than 1 bag of chips).
- *onpromotion* is the total number of items in a product group that were on promotion in the store on a given date.

The `stores.csv` file contains store metadata, including city, state, type, and cluster (all stores were grouped by similarity). The `oil.csv` file contains daily oil prices, covering both the training and the test periods. (Ecuador is an oil-dependent country, and its economic condition is strongly affected by oil-price fluctuations.)

2.2 Models and their validation

Three models were considered for forecasting the time series:

- ARIMA model with manual tuning of the parameters p , d , and q
- SARIMAX model
- Machine-learning model (gradient boosting, `XGBRegressor`)

Quality was assessed using the following metrics: MSE, RMSE, MAE, and absolute percentage error. The results were compared against a naive forecast — the average number of sales over the last 3 months.

For the SARIMAX and boosting models, two approaches were considered. In the first case, the model was fed time-based features: the index of the month, quarter, day of the year, and week, as well as the

day of the week. In addition, “lag features” were used: the number of sales on the same date one, two, and three years earlier (where data for that period exist). This approach allows forecasting any number of days ahead. In the second case, dataset-derived variables were added to these features: the fuel price on the chosen date, the number of promotions in stores, and the number of transactions. The difficulty of the second approach is that we do not know the future values of the added drivers in advance. To forecast dates not present in our dataset, the driver values must also be predicted.

2.3 Data preprocessing

The dataset variables required preprocessing. For example, the oil-price values had gaps, which were filled using quadratic-spline interpolation. Figures ?? and ?? show the price before and after interpolation, respectively.

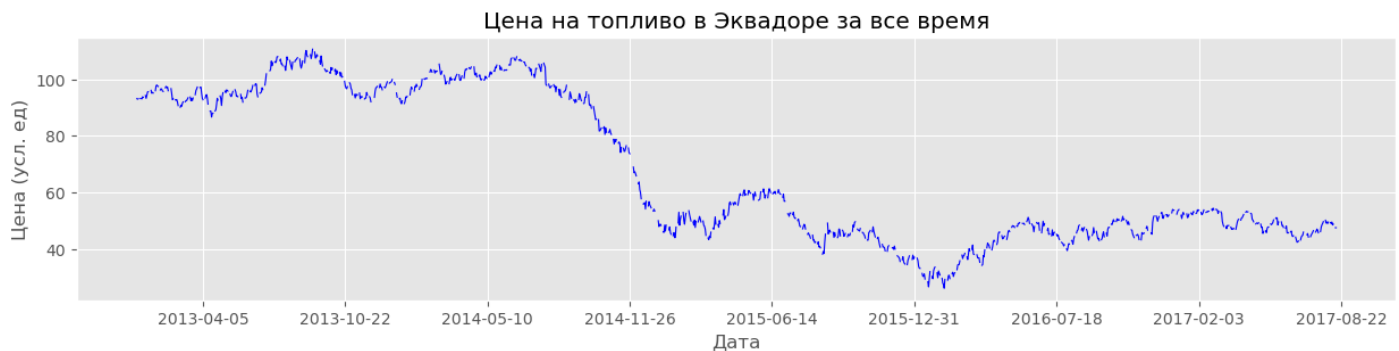


Figure 1: Fuel price per liter in Ecuador before interpolation.

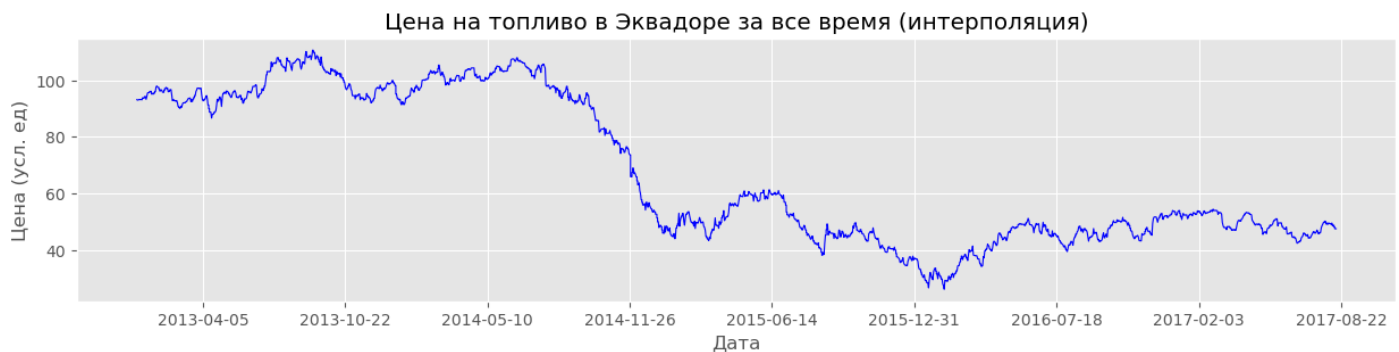


Figure 2: Fuel price per liter in Ecuador after interpolation.

In addition, categorical variables such as store type, store location, and the holiday flag were encoded numerically. However, all of these features were later excluded from the model due to low variance (a

feature could be filled entirely with a single value) or multicollinearity. Moreover, these features had a large share of missing values, on the order of 80 percent.

2.4 Model training

2.4.1 Naive forecast

As described earlier, the naive forecast used the average sales in Ecuador over the last 3 months of 2016. The forecast is shown in Figure ??.

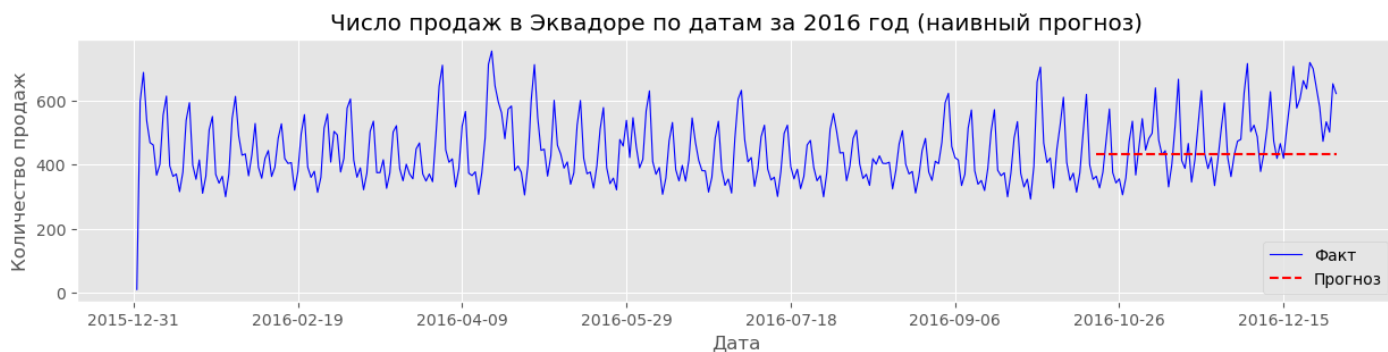


Figure 3: Naive forecast for sales in Ecuador.

The main metrics were measured for the forecast. Their values are given in Table ??.

Table 1: Metric values for the naive forecast.

	MSE	RMSE	MAE	Percentage error
Time features	20703.85	143.88	114.75	0.2

2.4.2 ARIMA model

The parameters for the ARIMA model were selected based on the autocorrelation and partial-autocorrelation functions. Stationarity of the series was tested using the Dickey–Fuller test.

It was found that the original series is not stationary (p -value : 0.68). After a single differencing, the series became stationary (p -value : $6.33 \cdot 10^{-14}$). In addition, a distribution plot of the residuals was produced; it turned out to be close to normal.

The results of the ARIMA model are shown in Figure ??.

The plot shows that the model does not capture the dynamics of the series, and the forecast quickly becomes constant. The metric values for the ARIMA model are given in Table ??.



Figure 4: ARIMA model forecast for the number of sales in Ecuador.

Table 2: Metric values for the ARIMA model forecast.

	MSE	RMSE	MAE	Percentage error
Time features	11908.6	109.13	93.53	1.491

2.4.3 SARIMAX model

The main difference of the SARIMAX model from the ARIMA model is its ability to account for seasonality in the prediction, as well as additional parameters beyond the values of the series itself. As mentioned above, the additional parameters fed to the model were the number of sales on the date, the fuel price, and the number of promotions in stores. The model was trained automatically using the Python *statsmodels* package. The model was trained twice: first on time and lag features only, and second with the additional variables. The model predictions are shown in Figure ???. The metric values are presented in Table ???.



Figure 5: SARIMAX model forecast for the number of sales in Ecuador.

The plot and metric values show that including additional parameters in the model gives a significant

Table 3: Metric values for the SARIMAX model forecast.

	MSE	RMSE	MAE	Percentage error
Time features	24258.07	155.75	125.47	0.22
Time features + additional	19785.57	140.661	88.76	0.15

improvement in prediction quality.

2.4.4 Boosting

As an extension to the work, it was of interest to see how boosting performs on the same input data. The working principle of boosting is fundamentally different from the econometric models considered above.

The work used XGBRegressor from the *sklearn* library. As with the SARIMAX model, training was carried out using two approaches. Hyperparameter search was performed with the *hyperopt* library. Unlike an ordinary grid search, this approach uses Bayesian optimization. In addition, the hyperparameter search used cross-validation adapted to time series.

The model predictions are shown in Figure ???. The metric values are presented in Table ???.



Figure 6: Gradient-boosting model forecast for the number of sales in Ecuador.

The plot and metric values show that for the specification with additional variables, the forecast accuracy is close to the actual sales values.

Table 4: Metric values for the gradient-boosting model forecast.

	MSE	RMSE	MAE	Percentage error
Time features	14971.71	122.36	86.83	0.15
Time features + additional	2745.55	52.4	39.19	0.07

3 Conclusion

This work solved the problem of forecasting the number of sales for a retail chain in Ecuador for 2016. Three forecasting approaches were considered:

- ARIMA model with manual tuning of the parameters p , d , and q
- SARIMAX model
- Machine-learning model (gradient boosting, XGBRegressor)

It was found that, among the econometric models, SARIMAX performed best. This result is due in particular to its ability to account for external parameters, rather than only the past values of the series.

In addition, the work considered forecasting with gradient boosting. This method outperformed all econometric models in prediction quality.